



JABchem



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Past Papers Higher Chemistry

2026 Marking Scheme

Grade Awarded	Mark Required		% candidates achieving grade
	(/150)	%	
A	105+	70%	%
B	87+	58%	%
C	70+	47%	%
D	52+	35%	%
No award	<52	<35%	%

Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	16.7 /25	53.5 /95	19.7 /30

2026 Higher Chemistry Marking Scheme

MC Qu	Answer	% Pupils Correct	Reasoning																														
1	D	62	☒ A Fullerene is a molecule of C_{60} . Ionic compounds are metal compounds. ☒ B Fullerene is a molecule of C_{60} . No metals in the structure to be metallic. ☒ C Fullerene is a solid with lower melting point, too low to be covalent network. ☒ D Fullerene is a discrete covalent molecule with a C_{60} structure.																														
2	C	81	☒ A H_2 has no hydrogen bonding as it has no O-H, N-H or F-H bonds. ☒ B There is no difference in electronegativity in H_2 molecule so H_2 is non-polar. ☒ C No electronegativity difference in an element $\therefore$ pure covalent bonding ☒ D Van der Waals forces are between molecules not inside molecules.																														
3	B	77	<table><tr><th>Answer</th><th>Particle</th><th>Size</th><th>Comparison</th><th>Size</th><th>Particle</th></tr><tr><td>☒ A</td><td>oxygen atom O</td><td>64</td><td><</td><td>140</td><td>oxide ion O^{2-}</td></tr><tr><td>☒ B</td><td>magnesium atom Mg</td><td>140</td><td>></td><td>72</td><td>magnesium ion Mg^{2+}</td></tr><tr><td>☒ C</td><td>sulfide ion S^{2-}</td><td>184</td><td>></td><td>104</td><td>sulfur atom S</td></tr><tr><td>☒ D</td><td>calcium ion Ca^{2+}</td><td>100</td><td><</td><td>174</td><td>calcium atom Ca</td></tr></table>	Answer	Particle	Size	Comparison	Size	Particle	☒ A	oxygen atom O	64	<	140	oxide ion O^{2-}	☒ B	magnesium atom Mg	140	>	72	magnesium ion Mg^{2+}	☒ C	sulfide ion S^{2-}	184	>	104	sulfur atom S	☒ D	calcium ion Ca^{2+}	100	<	174	calcium atom Ca
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5	A	48	☒ A Redox: metal + acid $\rightarrow$ salt + hydrogen ☒ B Neutralisation: metal hydroxide + acid $\rightarrow$ salt + water ☒ C Neutralisation: metal oxide + acid $\rightarrow$ salt + water ☒ D Neutralisation: metal carbonate + acid $\rightarrow$ salt + water + carbon dioxide																														
6	D	62	$2I^{-} \rightarrow I_2 + 2e^{-}$ is an oxidation reaction. As I^{-} is on right hand side of Electrochemical series (ECS), it will only react with a species below I^{-} on the left hand side of the electrochemical series. ☒ A SO_3^{2-} is on right side of ECS and, does not react with I^{-} ☒ B SO_4^{2-} is on left side of ECS but will not react with I^{-} as SO_4^{2-} is above I^{-} ☒ C Cr^{3+} is on the right side of ECS and does not react with I^{-} ☒ D $Cr_2O_7^{2-}$ is on the left side of ECS and below I^{-} redox reaction will proceed																														
7	B	74	☒ A oxygen in air causes the oxidation of food not the reduction. ☒ B oxygen can react with edible oils to form rancid products ☒ C hydrolysis of oil molecules would form glycerol and three fatty acid molecules ☒ D oil molecules are already unsaturated																														
8	B	71	☒ A ester shown is made from propan-1-ol and methanoic acid ☒ B ester shown is made from ethanol and propanoic acid ☒ C ester shown is made from ethanol and methanoic acid ☒ D ester shown is made from propan-1-ol and ethanoic acid																														
9	D	67	☒ A esters must have a minimum of 2 carbons e.g. methyl methanoate ☒ B alcohols have -OH groups but CH_2O_2 would have 2x-OH groups but not enough bonds to complete carbon ☒ C aldehydes contain only one oxygen atom e.g. methanal CH_2O ☒ D carboxylic acids have -COOH group e.g. methanoic acid $HCOOH$ (or CH_2O_2)																														
10	D	70	☒ A Addition: molecule added across $C=C$ double bond ☒ B Condensation: molecules joining together with water removed at the join ☒ C Hydrolysis: molecule split into smaller molecule with water added at break ☒ D Reduction: reverse reaction to oxidation carboxylic acid $\rightarrow$ ethanal is reduction																														

11	C	69	
12	A	86	☒ A fats are more saturated which allows molecules closer together and raises mpt ☒ B oils are less saturated than fats as they have more C=C double bonds ☒ C fats molecules pack together more closely as chains are straighter ☒ D oils molecules packed less closely together as oil molecules chain change direction at C=C double bonds
13	A	55	$\begin{array}{l} \textcircled{1} \quad \text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \longrightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O} \\ \textcircled{2} \quad \text{C}_3\text{H}_6\text{O} + \text{H}_2\text{O} \longrightarrow \text{C}_3\text{H}_6\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \end{array}$ $\begin{array}{l} \textcircled{1} \quad \text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \longrightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O} \\ \textcircled{2} \times 3 \quad 3\text{C}_3\text{H}_6\text{O} + 3\text{H}_2\text{O} \longrightarrow 3\text{C}_3\text{H}_6\text{O}_2 + 6\text{H}^+ + 6\text{e}^- \end{array}$ $\begin{array}{ccccccc} \text{Cr}_2\text{O}_7^{2-} & + & 8\text{H}^+ & + & 3\text{C}_3\text{H}_6\text{O} & \longrightarrow & 3\text{C}_3\text{H}_6\text{O}_2 + 2\text{Cr}^{3+} + 4\text{H}_2\text{O} \\ 1\text{mol} & & & & 3\text{mol} & & \\ \frac{1}{3}\text{mol} & & & & 1\text{mol} & & \end{array}$
14	B	57	☒ A 2,2-dimethylpentan-1-ol: same $\text{C}_7\text{H}_{16}\text{O}$ structure $\therefore$ not an isomer ☒ B 2,3,3-trimethylbutan-1-ol: different structure but same $\text{C}_7\text{H}_{16}\text{O}$ formula ☒ C octan-1-ol has 8 carbons so cannot be an isomer ☒ D 3,3,4-trimethylpentan-1-ol has 8 carbons so cannot be an isomer
15	D	47	☒ A molecule shown is glycerol (propane-1,2,3-triol) ☒ B molecule drawn is a fat/oil made by condensation of edible oil and glycerol ☒ C molecule shown is an ester formed by condensation of edible oil and an alcohol ☒ D an emulsifier is formed by the condensation of glycerol with 1 or 2 edible oil
16	C	49	☒ A $\text{S}_2\text{O}_8^{2-}$ is used up in Step 1 and not reformed later ☒ B I^- is used up in Step 2 and is not a catalyst ☒ C Fe^{2+} is catalyst as <i>not</i> used up in reaction (used up in step 1 and reformed in step 2) ☒ D SO_4^{2-} is formed in Step 2 and is not a catalyst
17	B	41	$\begin{array}{ccccccc} \text{Zn} & + & 2\text{AgNO}_3 & \longrightarrow & \text{Zn(NO}_3)_2 & + & 2\text{Ag} \\ 1\text{mol} & & 2\text{mol} & & 1\text{mol} & & 2\text{mol} \\ 0.025\text{mol} & & 0.05\text{mol} & & 0.025\text{mol} & & 0.05\text{mol} \\ (0.075\text{mol left over}) & & & & & & \end{array}$ ☒ A 0.025mol Zn reacts with 0.75mol Zn left over ☒ B all 0.05mol AgNO_3 reacts forming 0.05mol Ag ☒ C all 0.05mol of AgNO_3 reacts so is the limiting reactant and not in excess ☒ D 0.025mol Zn reacts with 0.75mol Zn left over
18	A	45	☒ A Fe^{3+} ions added would shift equilibrium to right forming more red colour ☒ B $[\text{FeCNS}]^{2+}$ added would shift equilibrium to left reducing red colour ☒ C NaCl added does not shift the equilibrium as Na^+ ions and Cl^- ions have no effect ☒ D Fe^{2+} ion is <i>not</i> a reactant or product so no effect on position of equilibrium
19	D	85	☒ A Decreased activation energy would make successful collisions more likely ☒ B Increased kinetic energy of reactants would increase the rate of reaction ☒ C Increased concentration of reactants would increase rate of reaction ☒ D the enthalpy change is independent of the rate of reaction

20	B	81	☒ A distillation is the separation of substances with different boiling points ☒ B the polarity of a substance is used to separate chemicals by chromatography ☒ C reactivity involves chemical reactions but chromatography separates by variations in physical properties like size and polarity/solubility in mobile phase ☒ D chromatography separates substances due to size & polarity regardless of colour																				
21	A	31	☒ A one spot in different location from X and Y indicating pure substance is formed ☒ B both spots correspond to either original X or original Y - no new substance ☒ C new substance and X in sample. Not a pure sample of fragrance produced ☒ D two new substances produced not corresponding to either original X or Y																				
22	C	80	☒ A no results within 0.2cm^3 of each other ☒ B all are 0.2cm^3 away from another experiment but all experiments must be within 0.2cm^3 ☒ C all results within 0.2cm^3 range ☒ D 2 results within 0.2cm^3 of each other but experiment 2 more than 0.2cm^3 different																				
23	C	56	☒ A droppers do not give accurate measurements ☒ B conical flasks do not give accurate measurements ☒ C pipette volumes and volumetric flasks is most accurate way to prepare solutions ☒ D 50cm^3 measured from a burette not as accurate as 50cm^3 measured in a pipette																				
24	D	68	☒ A Analysing 1,4,6 may give information about position of sidechain and octane no. ☒ B no conclusion can be drawn from analysing 1,2,4 ☒ C no conclusion can be drawn from analysing 3,4,5 ☒ D 2,3,5 demonstrate the effect of increasing the length of straight chain																				
25	C	57	<table border="1"> <thead> <tr> <th>Liquid</th><th>X</th><th>Y</th><th>Z</th></tr> </thead> <tbody> <tr> <td>Name of Liquid</td><td>glycerol</td><td>propanone</td><td>ethanol</td></tr> <tr> <td>Intermolecular Bonding Type</td><td>3x hydrogen bonding</td><td>permanent dipole - permanent dipole attractions</td><td>1x hydrogen bonding</td></tr> <tr> <td>Explanation of speed of ball falling</td><td>High degree of hydrogen bonding results in molecules close together and ball slowest travelling through X</td><td>Propanone molecules not as close together due to weaker permanent dipole to permanent dipole attractions between molecules. Ball travels fastest through Y</td><td>Lower degree of hydrogen bonding results in molecules not as close together as X but still close together. Ball travels slower through Z</td></tr> <tr> <td>Conclusion</td><td>Slowest</td><td>Fastest</td><td>Middle value</td></tr> </tbody> </table>	Liquid	X	Y	Z	Name of Liquid	glycerol	propanone	ethanol	Intermolecular Bonding Type	3x hydrogen bonding	permanent dipole - permanent dipole attractions	1x hydrogen bonding	Explanation of speed of ball falling	High degree of hydrogen bonding results in molecules close together and ball slowest travelling through X	Propanone molecules not as close together due to weaker permanent dipole to permanent dipole attractions between molecules. Ball travels fastest through Y	Lower degree of hydrogen bonding results in molecules not as close together as X but still close together. Ball travels slower through Z	Conclusion	Slowest	Fastest	Middle value
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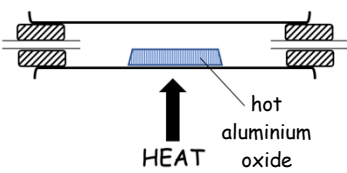
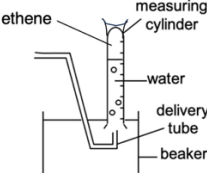
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Long Qu	Answer	Reasoning														
1a(i)	lattice	An ionic lattice is a 3-dimensional structure of alternating positive ions and negative ions held tightly together by electrostatic attraction														
1a(ii)A	attraction between positive and negative (ions)	Electrostatic attraction is the attraction between a positive particle and a negative particle. Opposite charges are attracted to each other. The same charge will repel each other.														
1a(ii)B	large difference in electronegativity	<table><tr><td>Bonding Type</td><td>Non-polar Covalent</td><td>Polar Covalent</td><td>Ionic Bonding</td></tr><tr><td>Difference in Electronegative</td><td>Small difference or zero</td><td>—————></td><td>Large Difference</td></tr></table>	Bonding Type	Non-polar Covalent	Polar Covalent	Ionic Bonding	Difference in Electronegative	Small difference or zero	—————>	Large Difference						
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1b(i)	3	Lithium sulfate has a formula of Li ₂ SO ₄ and an ionic formula (Li ⁺) ₂ SO ₄ ²⁻ Lithium sulfate has three ions per formula ion (2xLi ⁺ and 1xSO ₄ ²⁻)														
1b(ii)	100.7	$\begin{aligned}\Delta T_b &= 0.51 \times c \times i \\ \Delta T_b &= 0.51 \times 0.35 \times 4 \\ \Delta T_b &= 0.71^{\circ}\text{C} \\ \text{Boiling Point} &= 100^{\circ}\text{C} + 0.71^{\circ}\text{C} = 100.71^{\circ}\text{C}\end{aligned}$														
2a	Answer to include:	<table><tr><td>1st Mark:</td><td colspan="2">description of LDFs as 2 forces of attraction between temporary dipoles (and induced dipoles)</td></tr><tr><td>2nd Mark:</td><td>explanation of the cause of temporary dipoles in terms of</td><td> uneven distribution of electrons electron wobble movement of electrons in the molecule </td></tr></table>	1 st Mark:	description of LDFs as 2 forces of attraction between temporary dipoles (and induced dipoles)		2 nd Mark:	explanation of the cause of temporary dipoles in terms of	uneven distribution of electrons electron wobble movement of electrons in the molecule								
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2b	They have the same number of electrons <u>or</u> strength of LDF is same	London Dispersion forces are decided by the number of electrons a molecule has. H ₂ O has 10 electrons (8 from O atom and 1 each from 2 hydrogens) CH ₄ has 10 electrons (6 from C atom and 1 each from 4 hydrogens)														
2c	Answer to include:	<table><tr><td>1st Mark:</td><td>Water contains polar bonds</td></tr><tr><td>2nd Mark:</td><td>Has overall dipole <u>or</u> shape of molecule means dipoles do not cancel out.</td></tr></table>	1 st Mark:	Water contains polar bonds	2 nd Mark:	Has overall dipole <u>or</u> shape of molecule means dipoles do not cancel out.										
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3	Open Question: <table><tr><td>0 marks</td><td>1 mark</td><td>2 marks</td><td>3 marks</td></tr><tr><td>11%</td><td>38%</td><td>42%</td><td>9%</td></tr></table>	0 marks	1 mark	2 marks	3 marks	11%	38%	42%	9%	<table><tr><th>3 mark answer</th><th>2 mark answer</th><th>1 mark answer</th></tr><tr><td>Demonstrates a <u>good</u> understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.</td><td>Demonstrates a <u>reasonable</u> understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td>Demonstrates a <u>limited</u> understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.</td></tr></table>	3 mark answer	2 mark answer	1 mark answer	Demonstrates a <u>good</u> understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a <u>reasonable</u> understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited</u> understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.
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4a(i)	Line showing:	Line leaving R going up upwards, peaking and coming down Line descending and levelling off at P. Distance P is below R should be greater that distance peak/activated complex is above R														
4a(ii)	173	$\begin{aligned}\text{Activation energy for reverse reaction} &= \text{enthalpy change for reverse reaction} + \text{activation energy for forward reaction} \\ \text{Activation energy for reverse reaction} &= +98\text{kJ mol}^{-1} + 75\text{kJ mol}^{-1} \\ \text{Activation energy for reverse reaction} &= 173\text{ kJ mol}^{-1}\end{aligned}$														
4b	Answer to include	<table><tr><td>1st Mark:</td><td>Catalyst provides an alternative reaction pathway</td></tr><tr><td>2nd Mark:</td><td>Lower activation energy or More particles have energy greater than activation energy</td></tr></table>	1 st Mark:	Catalyst provides an alternative reaction pathway	2 nd Mark:	Lower activation energy or More particles have energy greater than activation energy										
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4c	0.17	$\text{no. of moles} = \frac{\text{Volume}}{\text{Molar Volume}} = \frac{0.060\text{litres}}{24\text{ litres mol}^{-1}} = 0.0025\text{mol}$ $\begin{array}{ccc} 2\text{H}_2\text{O}_2 & \longrightarrow & 2\text{H}_2\text{O} + \text{O}_2 \\ 2\text{mol} & & 2\text{mol} \quad 1\text{mol} \\ 0.0050\text{mol} & & 0.0025\text{mol} \end{array}$ gfm H₂O₂ = (2x1)+(2x16) = 2+32 = 34g mass = no. of mol x gfm = 0.0050mol x 34 = 0.17g														

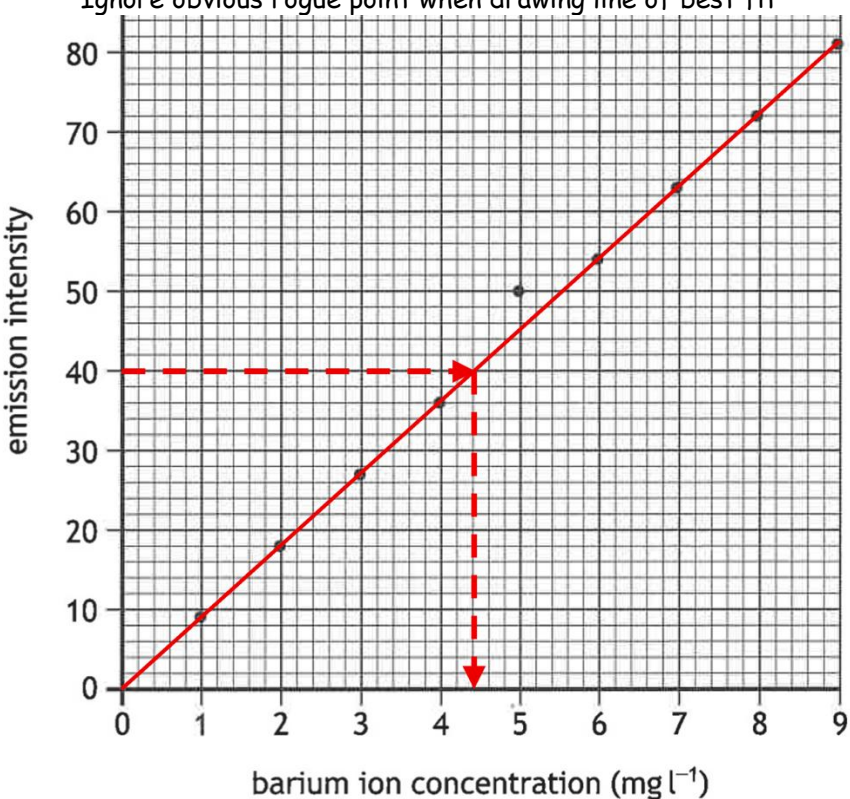
4d(i)	Atom/molecule with unpaired electron	Free radicals are particles with a unpaired/single electron. Free radicals are very reactive particles and usually generated by the presence of light.									
4d(ii)	HO•	If the oxygen to oxygen single bond is the weakest bond then the hydrogen peroxide molecule splits at the O-O single to form HO•. $\text{H}_2\text{O}_2 \longrightarrow 2\text{HO}^\bullet$									
4e(i)	(Number of) molecules with energy greater than the activation energy	The activation energy is the minimum energy required for a successful collision. Any particle with energy greater than the activation energy can react and become products assuming it collides at the correct angle.									
4e(ii)	Line drawn to the right of the original curve and not as high										
5a	volatile or non-water soluble	Essential oils are concentrated extracts of the volatile, non-water-soluble aroma compounds from plants. They are mixtures of many different compounds. They are widely used in perfumes, cosmetic products, cleaning products and as flavourings in foods.									
5b(i)		Isoprene is also known as 2-methylbuta-1,3-diene and has a molecular formula of C₅H₈	2-methylbuta-1,3-diene								
5b(ii)	2	Formula of limonene is C ₁₀ H ₁₆ . Limonene is made from two C ₅ H ₈ units									
5c(i)	Hexane is non-polar. Limonene/oils are non-polar	Hexane is a non-polar solvent and would dissolve essential oils as essential oils are non-polar themselves. Water is a polar solvent and would not dissolve non-polar essential oils.									
5c(ii)	-4547.84	$\begin{aligned} \text{Heat Energy} &= \text{Specific Heat Capacity} \times \text{Mass} \times \text{Change In Temperature} \\ E_h &= c \times m \times \Delta T \\ E_h &= 4.18 \text{ kJ kg}^{-1} \text{ }^\circ\text{C}^{-1} \times 0.5\text{kg} \times 56^\circ\text{C} \\ E_h &= 117.04 \text{ kJ} \end{aligned}$ gfm Limonene C₇H₁₆ = 136g $\begin{aligned} 3.5\text{g limonene} &\longleftrightarrow 117.04\text{kJ} \\ 136\text{g limonene} &\longleftrightarrow 117.04\text{kJ} \times \frac{136}{3.5} \\ &= -4547.84 \text{ kJ mol}^{-1} \end{aligned}$									
5d(i)	2 carbons attached to the carbon with to the -OH group or 1 hydrogen attached to the carbon with to the -OH group	<table><thead><tr><th>Alcohol Type</th><th>Definition</th></tr></thead><tbody><tr><td>Primary</td><td> 1 carbon 2 hydrogens attached to the carbon with the -OH group</td></tr><tr><td>Secondary</td><td> 2 carbons 1 hydrogen attached to the carbon with the -OH group</td></tr><tr><td>Tertiary</td><td> 3 carbons 0 hydrogens attached to the carbon with to the -OH group</td></tr></tbody></table>		Alcohol Type	Definition	Primary	1 carbon 2 hydrogens attached to the carbon with the -OH group	Secondary	2 carbons 1 hydrogen attached to the carbon with the -OH group	Tertiary	3 carbons 0 hydrogens attached to the carbon with to the -OH group
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5d(ii)		Secondary alcohols are oxidised to ketone groups by oxidising agents. Groups containing the carbonyl group are: <table><tr><td></td><td></td><td></td><td></td></tr><tr><td>carbonyl group</td><td>carboxyl group</td><td>aldehyde group</td><td>ketone group</td></tr></table>					carbonyl group	carboxyl group	aldehyde group	ketone group							
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6a(i)	3-methylbutan-1-ol																
6a(ii)	Brown (solid) formed	<table><tr><td>Oxidising agent</td><td>Start Colour</td><td>End colour</td></tr><tr><td>Hot copper(II) oxide</td><td>black</td><td>brown</td></tr><tr><td>Acidified dichromate solution</td><td>orange</td><td>green</td></tr><tr><td>Fehling's solution</td><td>blue</td><td>brick red</td></tr><tr><td>Tollen's Reagent</td><td>(colourless)</td><td>silver mirror formed</td></tr></table>	Oxidising agent	Start Colour	End colour	Hot copper(II) oxide	black	brown	Acidified dichromate solution	orange	green	Fehling's solution	blue	brick red	Tollen's Reagent	(colourless)	silver mirror formed
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6b	1315	Total volume = 27000 litres ∴ 4% of 27000 litres = $\frac{4}{100} \times 27000 = 1080$ litres no. of moles = $\frac{\text{Volume}}{\text{Molar Volume}} = \frac{1080 \text{ litres}}{23 \text{ litres mol}^{-1}} = 46.96 \text{ mol}$ mass = no. of mol x gfm = $46.96 \times 28 = 1314.78\text{g}$															
6c(i)	proteins	Enzymes are biological catalysts made from specially shaped proteins.															
6c(ii)	Answer to include:	<table><tr><td>1st Mark:</td><td>intermolecular bonds van der Waals forces hydrogen bonds</td><td rowspan="2">} (within protein) break</td></tr><tr><td>2nd Mark:</td><td>Protein changes shape or chains unwind</td></tr></table>	1 st Mark:	intermolecular bonds van der Waals forces hydrogen bonds	} (within protein) break	2 nd Mark:	Protein changes shape or chains unwind										
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7a(i)	One arrow from:																

7a(ii)	distillation	Distillation separates liquids with different boiling points. Ethanol has b.pt. of 78°C and water has b.pt. of 100°C.																											
7a(iii)A	Addition	Addition reactions add a molecules across a C=C double bond or a C≡C triple bond. Addition of water is also called a hydration reaction.																											
7a(iii)B	Answer to include:	1 st Mark:	Increase in pressure favours side with fewest moles of gas																										
		2 nd Mark:	Rate of forward reaction increases or rate of forward reaction speeds up																										
7a(iii)C	-41	<table><thead><tr><th colspan="2">Bond Breaking Steps</th><th colspan="2">Bond Forming Steps</th></tr></thead><tbody><tr><td>1x C=C bond</td><td>1x 614kJ = 614kJ</td><td>1x C-C bond</td><td>1x 346kJ = 346kJ</td></tr><tr><td>4x C-H bonds</td><td>4x 412kJ = 1648kJ</td><td>5x C-H bonds</td><td>5x 412kJ = 2060kJ</td></tr><tr><td>2x O-H bonds</td><td>2x 463kJ = 926kJ</td><td>1x O-H bond</td><td>1x 463kJ = 463kJ</td></tr><tr><td></td><td></td><td>1x C-O bond</td><td>1x 360kJ = 360kJ</td></tr><tr><td>Total bond breaking</td><td>= 3188kJ</td><td>Total bond Forming</td><td>= 3229kJ</td></tr></tbody></table> $\Delta H = \Sigma \text{Bond enthalpies for bonds broken} - \Sigma \text{Bond enthalpies for bonds formed}$ $\Delta H = 3188 - 3229$ $\Delta H = -41 \text{ kJ mol}^{-1}$				Bond Breaking Steps		Bond Forming Steps		1x C=C bond	1x 614kJ = 614kJ	1x C-C bond	1x 346kJ = 346kJ	4x C-H bonds	4x 412kJ = 1648kJ	5x C-H bonds	5x 412kJ = 2060kJ	2x O-H bonds	2x 463kJ = 926kJ	1x O-H bond	1x 463kJ = 463kJ			1x C-O bond	1x 360kJ = 360kJ	Total bond breaking	= 3188kJ	Total bond Forming	= 3229kJ
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7a(iii)D	One answer from:	only one product or no waste products or no by-products	all atoms in reactants are converted into the products	mass of product is equal to mass of reactants	Proportion of total mass of all starting materials successfully converted into desired product is 100%																								
7b(i)	Increase in the oxygen to hydrogen ratio	Oxidation	Increase in the oxygen : hydrogen ratio																										
		Reduction	Decrease in the oxygen : hydrogen ratio																										
7b(ii)	Ethanoic acid	Primary Alcohol $\xrightarrow{\text{oxidation}}$ Aldehyde $\xrightarrow{\text{oxidation}}$ Carboxylic Acid Secondary Alcohol $\xrightarrow{\text{oxidation}}$ Ketone $\xrightarrow[\text{no oxidation}]{\times}$ Tertiary Alcohol $\xrightarrow[\text{no oxidation}]{\times}$																											
7b(iii)A	reactants/products are flammable	Ethanol is a flammable substance which boils at 78°C and ethanol vapour could catch fire if a Bunsen burner is used to heat the water.																											
7b(iii)B	orange to green	<table><thead><tr><th>Oxidising agent</th><th>Start Colour</th><th>End colour</th></tr></thead><tbody><tr><td>Hot copper(II) oxide</td><td>black</td><td>brown</td></tr><tr><td>Acidified dichromate solution</td><td>orange</td><td>green</td></tr><tr><td>Fehling's solution</td><td>blue</td><td>brick red</td></tr><tr><td>Tollen's Reagent</td><td>(colourless)</td><td>silver mirror formed</td></tr></tbody></table>				Oxidising agent	Start Colour	End colour	Hot copper(II) oxide	black	brown	Acidified dichromate solution	orange	green	Fehling's solution	blue	brick red	Tollen's Reagent	(colourless)	silver mirror formed									
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7b(iii)C	$\begin{array}{c} \text{C}_2\text{H}_5\text{OH} + \text{H}_2\text{O} \\ \downarrow \\ \text{CH}_3\text{COOH} + 4\text{H}^+ + 4\text{e}^- \end{array}$	<u>Step 1</u> Write down main species $\text{C}_2\text{H}_5\text{OH} \longrightarrow \text{CH}_3\text{COOH}$ <u>Step 2</u> Balance atoms other than O and H $\text{C}_2\text{H}_5\text{OH} \longrightarrow \text{CH}_3\text{COOH}$ <u>Step 3</u> Add H₂O to other side to balance O $\text{C}_2\text{H}_5\text{OH} + \text{H}_2\text{O} \longrightarrow \text{CH}_3\text{COOH}$ <u>Step 4</u> Add H⁺ to other side to balance H $\text{C}_2\text{H}_5\text{OH} + \text{H}_2\text{O} \longrightarrow \text{CH}_3\text{COOH} + 4\text{H}^+$ <u>Step 5</u> Add e⁻ to most positive side to balance charge $\text{C}_2\text{H}_5\text{OH} + \text{H}_2\text{O} \longrightarrow \text{CH}_3\text{COOH} + 4\text{H}^+ + 4\text{e}^-$																											
7b(iii)D	acids supply H ⁺ ions (for reaction)	If any of the reactants are missing the reaction cannot proceed to form products. Acids are used to acidify reactions to provide H ⁺ ions.																											
7b(iii)E	tertiary	Oxidising Agent	primary alcohol ↓ aldehyde	secondary alcohol ↓ ketone	aldehyde ↓ carboxylic acid																								
		acidified dichromate	✓	✓	✓																								
7b(iv)	Fehling's Solution or Tollen's Reagent	hot copper(II) oxide	✓	✓	✓																								
		Fehling's solution	x	x	✓																								
		Tollen's reagent	x	x	✓																								

7c(i)	Diagram showing:	1 mark 	1 mark 		
7c(ii)	unsaturated	Unsaturated compounds are decolourised by bromine solution quickly. Bromine adds across the C=C double bond to leave a single bond with a bromine atom on either side of the that bond.			
8a(i)	condensation	Condensation polymerisation occurs when monomers join together to make a polymer with small molecules e.g. water removes at the joins			
8a(ii)	One dipeptide from:	O H H O H HO—C—C—N—C—C—NH₂ CH₂ H₃C—CH SH CH₂ CH₃ or O H H O H HO—C—C—N—C—C—NH₂ H₃C—CH CH₂ CH₂ SH CH₃			
8b(i)A	hydrophobic	acidic	basic	hydrophobic	polar
8b(i)B	sidechain contains carboxyl or COOH group	Acidic side chains usually contain a carboxyl group e.g. -COOH	Basic side chains usually contain. A amino group e.g. -NH ₂	Hydrophobic side chains contain all non-polar hydrocarbon groups	Polar side groups will contain groups with potential for hydrogen bonding e.g. hydroxyl -OH groups
8b(ii)	Hydrogen bonding	Hydrogen bonding occurs between molecules containing -OH, -NH and H-F bonds			
8c	Strong covalent bonds must be broken	Proteins can be denatured when proteins change their shape as heat overcomes the strength of hydrogen bonding between side chains in the protein. Disulfide bridges contain covalent bonds that need to be broken before denaturing can take place.			
9	Open Question:	3 mark answer		2 mark answer	1 mark answer
		Demonstrates a <u>good</u> understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.		Demonstrates a <u>reasonable</u> understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited</u> understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.
10a(i)	Decrease Decrease	Decrease in Temperature		Decrease in concentration of NH ₃	
		Decreasing in temperature will favour the forward exothermic reaction which increases the yield of ammonia NH ₃		Decreasing the concentration of NH ₃ will shift the equilibrium to the right to replace the missing NH ₃	
10a(ii)	No effect	Catalysts lower the activation energy but have no effect on the final position of equilibrium. This means the equilibrium concentration of NH ₃ is the same.			
10a(iii)	66.7%	no. of mol = $\frac{\text{mass}}{\text{gfm}}$ = $\frac{500000}{28}$ = 17857mol N₂ + 3H₂ —————→ 2NH₃ 1mol 17857mol 2mol 35714mol mass = no.of mol × gfm = 35714 × 17 = 607143g = 607kg (theoretical) %Yield = $\frac{\text{Actual}}{\text{Theoretical}} \times 100$ = $\frac{405\text{kg}}{607\text{kg}} \times 100$ = 66.7%			

10b	206	① CO + $\frac{1}{2}$O₂ → CO₂ ΔH=-283 kJ mol⁻¹ ② H₂ + $\frac{1}{2}$O₂ → H₂O ΔH=-242 kJ mol⁻¹ ③ CH₄ + 2O₂ → CO₂ + 2H₂O ΔH=-803 kJ mol⁻¹ ①x-1 CO₂ → CO + $\frac{1}{2}$O₂ ΔH=+283 kJ ②x-3 $\frac{3}{2}$ H₂O → 3H₂ + $\frac{3}{2}$O₂ ΔH=+726 kJ ③ CH₄ + 2O₂ → CO₂ + 2H₂O ΔH=-803 kJ Add ①'+②+③ CH₄ + H₂O → CO + 3H₂ ΔH=+206 kJ mol⁻¹			
10c(i)	19.2%	atom economy = $\frac{\text{mass of desired product(s)}}{\text{total mass of reactants}} \times 100 = \frac{(1 \times 17)}{(1 \times 34.7) + (3 \times 18.0)} \times 100 = 19.2\%$			
10c(ii)	283.3	0.0013g ammonia ←→ 78s 1mol ammonia = 17g ammonia ←→ 78s x $\frac{17}{0.0013} = 1020000\text{s}$ time = $\frac{1020000\text{s}}{60 \text{ s/min}} = 17000\text{min} = \frac{17000\text{min}}{60\text{min/hour}} = 283.3 \text{ hours}$			
11a(i)	standard solution	Standard solutions are solutions of accurately known concentration			
11a(ii)	1.125	no. of mol = volume x concentration = 0.25 litres x 0.050 mol l⁻¹ = 0.0125mol gfm oxalic acid = 90g mass = no. of mol x gfm = 0.0125 x 90 = 1.125g			
11a(iii)	Two answers from:	1 mark each for any two answers of the following: <table><tr><td>Rinse beaker and transfer the rinsings/washings into the flask or transfer quantitatively into flask</td><td>Use deionised water (to make up to the mark)</td><td>make up so that bottom of meniscus meets the line on flask</td></tr></table>	Rinse beaker and transfer the rinsings/washings into the flask or transfer quantitatively into flask	Use deionised water (to make up to the mark)	make up so that bottom of meniscus meets the line on flask
Rinse beaker and transfer the rinsings/washings into the flask or transfer quantitatively into flask	Use deionised water (to make up to the mark)	make up so that bottom of meniscus meets the line on flask			
11b(i)	to determine the end-point of reaction	Indicators are used in titration to accurately detect when the endpoint of a reaction has been reached. (Permanganate is self-indicating)			
11b(ii)	One answer from:	<table><tr><td>tap opened solution drained</td><td>to ensure no air bubbles</td><td>solution run onto the scale</td></tr></table>	tap opened solution drained	to ensure no air bubbles	solution run onto the scale
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11b(iii)A	One answer from:	<table><tr><td>First result is not concordant</td><td>It is the average of concordant results</td><td>A description of concordancy with volumes within 0.2cm³</td></tr></table>	First result is not concordant	It is the average of concordant results	A description of concordancy with volumes within 0.2cm ³
First result is not concordant	It is the average of concordant results	A description of concordancy with volumes within 0.2cm ³			
11b(iii)B	0.111	no. of mol = volume x concentration = 0.02695 litres x 0.0517 mol l⁻¹ = 0.00139 mol (COOH)₂ 1mol 0.00139mol + 2NaOH 2mol 0.00279mol → 2H₂O + (COONa)₂ concentration = $\frac{\text{no. of mol}}{\text{volume}} = \frac{0.00279 \text{ mol}}{0.025 \text{ litres}} = 0.111 \text{ mol l}^{-1}$			
11b(iii)C	O Na⁺-O-C-C-O⁻-Na⁺ O	Carboxyl groups -COOH provide the hydrogen H ⁺ ions for reaction with bases like sodium hydroxide forming a salt and water.			
12a(i)	sodium and cadmium	<table><tr><td>Na</td><td> </td></tr></table>	Na		
Na					

12a(ii)	green	According to the colour wheel on page 19 of data booklet, the colour associated with 535nm is green.
12b(i)	4.4	Ignore obvious rogue point when drawing line of best fit  emission intensity barium ion concentration (mg l^{-1})
12b(ii)	hard (water)	Hard water contains high levels of dissolved metal ions, usually calcium and magnesium
12c(i)	$\text{He(g)} \rightarrow \text{He}^{\text{+}}(\text{g}) + \text{e}^{-}$	The first ionisation is the energy required to remove one mole of electrons from one mole of atoms in the gaseous state.
12c(ii)	Helium has only 2 electrons or No third electron in helium	Helium atoms have an electron arrangement of 2. The 1 st ionisation energy removes an electron to form the ion $\text{He}^{\text{+}}$ with an electron arrangement of 1. The second ionisation energy removes the second electron of $\text{He}^{\text{+}}$ (electron arrangement 1) to form the ion He^{2+} and has an electron arrangement of 0. There are no electrons left for a third ionisation to take place.